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# The Palaeomagnetism of the Samoan and Tongan Islands

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*Summary*

The remanent magnetization of the Samoan rocks is of varying stability. Samples from sixty sites are regarded as reliable indicators of the polarity of remanence but only sixteen sites are regarded as reliable indicators of the actual direction of the past geomagnetic field. Normal and reversed directions occur, the mean direction of which (5°, −32°) is similar to the axial dipole field in Samoa. The scatter of Samoan site directions about the mean is greater than would be expected from secular variation rates during the last 200 years and suggests that secular variation in the Samoan area has not been abnormal when averaged over the time taken for the rocks to be laid down (2–2·5 m.y.). Samples from Tonga are stable but are insufficient in number for detailed interpretation, but results from 'Eua suggest tectonic instability on the edge of the Tonga Trench before the Upper Eocene.

## 1. Introduction

The palaeomagnetic properties of igneous rock samples from Samoa and Tonga were examined as part of an investigation of the variation of the geomagnetic field during the last few million years in islands in and around the margin of the Pacific Ocean. In particular it was hoped (a) to estimate the magnitude of secular variation in this region of slow change during historic times, (b) to test the hypothesis of world-wide reversals of the geomagnetic field and to use such reversals for geological correlation in an area where standard stratigraphical techniques are difficult to apply, and (c) to test the axial dipole hypothesis of the average geomagnetic field during the late Cainozoic.

Palaeomagnetic results from the Hawaiian Islands have been described earlier (Doell & Cox 1961, 1963, Tarling 1965) and the presence of reversals of magnetization in Samoa has been noted previously (Tarling 1962). The Samoan and Tongan Islands are in close proximity but these islands are representative of two distinct geological environments. The Samoan volcanics form part of the oceanic basalt suite of rocks while the Tongan Islands, lying along the Andesite Line, are representative of a 'continental' or 'subcontinental' environment. Therefore each island group is described separately following an outline of the methods used.

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## 2. Methods

Oriented hand samples of igneous rocks were collected from various sites in both groups of islands. Almost all sites, which are single lava flows or dykes, were natural exposures, mostly along the base of ancient sea cliffs in Eastern Samoa and the present-day sea cliffs in the Tongan Islands. Two separate specimens were drilled in the laboratory from most samples and the intensity and direction of the natural remanent magnetization (NRM) of these specimens were measured with astatic magnetometers (Irving 1956) with an accuracy of 3° in direction and 5% in intensity. The stability of NRM was studied by the alternating magnetic field method (Thellier & Rimbert 1954) using the apparatus and techniques described by Irving *et al.* (1961a). Low coercivity components were removed from all Samoan specimens by treatment in 150 oe (peak) alternating magnetic field and from all Tongan specimens in 225 oe (peak) alternating field, and the remaining intensity and direction was remeasured.

The directions of NRM, before and after treatment, were analysed statistically to determine average directions and to estimate the scatter of individual directions about the mean. Specimen directions were averaged graphically, giving equal weight to each specimen, to obtain sample directions, and the sample directions at each site were then added vectorally, giving equal weight to each sample, to obtain resultant site vector (**R**). The significance of site directions was tested by comparing the length of **R** against the vector length, corresponding to 95% significance (Watson 1956a, Vincenz & Bruckshaw 1960, M. A. Ward, personal communication). As the sites were chosen so that samples from the same site can be assumed to have acquired their thermoremanence at the same time, stable samples from the same site should have, within experimental error, the same directions of remanence. Therefore sites at which **R** was not significant at a 95% level after partial demagnetization were regarded as unreliable indicators of the precise direction of the past geomagnetic field, although the specimen directions, if all near the normal or reversed axial dipole field, are thought to be a valid indication of the polarity of the ambient field when the rocks cooled. Significant site directions were combined vectorally to obtain the overall mean direction.

The number of samples available at each site renders the estimates of precision, etc., unreliable as only two or three samples were obtained at most sites and the probable error of the estimate increases rapidly as the number of samples per site falls below five. However the directions of NRM in Samoa were scattered and some objective method must be applied to the data to distinguish results on which more reliance can be placed. It was considered that the use of the 95% significance level affords a means of doing this although the method was not completely satisfactory. It was possible that equal weight could be given to each specimen in calculation of the precision estimates but it was considered that the more conservative approach of giving equal weight to each sample was a better procedure when the number of specimens per sample was variable and could yield an unrealistically high precision when a large number of specimens were obtained from a single sample.

The scatter of significant Samoan site directions about their mean was examined by two methods. The precision of the overall mean direction was estimated using the Fisher (1953) method, and the method given by Watson (1956b) and Watson & Irving (1957) was then used in which allowance is made for the effect of within-site scatter (between different samples at the same site) ($\omega$) on the between-site scatter ($\beta$).

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## 3. Samoa

The Samoan Islands are divided into two political units, American (Eastern) and Western Samoa (Fig. 1). American Samoa comprises the island of Tutuila (130 $km^2$) and several smaller volcanic and limestone islands, while Western Samoa consists of the islands of Upolu (1 000 $km^2$), Savai'i (1 800 $km^2$) and a few smaller volcanic and limestone islands. The volcanic islands are formed by olivine–basalt lavas, although more acidic and more basic rocks occur throughout the sequence. The paucity of sedimentary rocks and other marker horizons has made geological correlation (Table 1) difficult. The age is thought to decrease from Pliocene in the E to Recent in the W—a decrease in age in the opposite direction to all other chains of subaerial volcanic islands within the Pacific Ocean (Chubb 1957).

**Table 1**

*Igneous formations in Samoa, based on Stearns (1944) and Kear & Wood (1959)*

| Geological Period | Tutuila | Upolu | Savai'i | Polarity | Age* |
|---|---|---|---|---|---|
| Historic | | | Aopo 1905–11 | N | |
| | | | Aopo *c.* 1760 | N | |
| | Leone† | Puapua | Puapua | N | |
| Post-Glacial | | Lefaga | | N | |
| Last Glacial | | Mulifanua | Mulifanua | N | |
| Last Inter-glacial | | Salani | Salani | N | |
| | | Upper Fagaloa‡ | Fagaloa‡ | N | 1 m.y. |
| | | Lower Fagaloa‡ | | R | |
| | Taputapu‡ | | | R | |
| Plio-Pleistocene | Pago Intracaldera | | | R | |
| | Pago Extracaldera | | | R | |
| | Alofau | | | R | |
| | Olomoana | | | R | <2·6 m.y. |
| | Masefau Dyke Complex | | | ? | |

* Based on comparison with Hawaiian lavas (Tarling 1962, McDougall & Tarling 1963).
† Correlation based on morphology, etc., by Tarling (this paper).
‡ Stratigraphical relationship based on palaeomagnetic evidence of reversals.

A total of 110 hand samples were obtained from sixty sites in all the volcanic formations except one on the three largest islands. One site was in a dyke, but all other sites were in separate lava flows. Only one sample could be obtained at thirty-five sites, and of the remaining twenty-five sites where the significance of the mean direction could be tested (section 2), sixteen had significant directions after partial demagnetization, in 150 oe (peak) alternating field. The stability of NRM is not high when tested in alternating magnetic fields, but it is considered that samples from the sixteen significant sites are reliable indicators of the actual direction of the past geomagnetic field.



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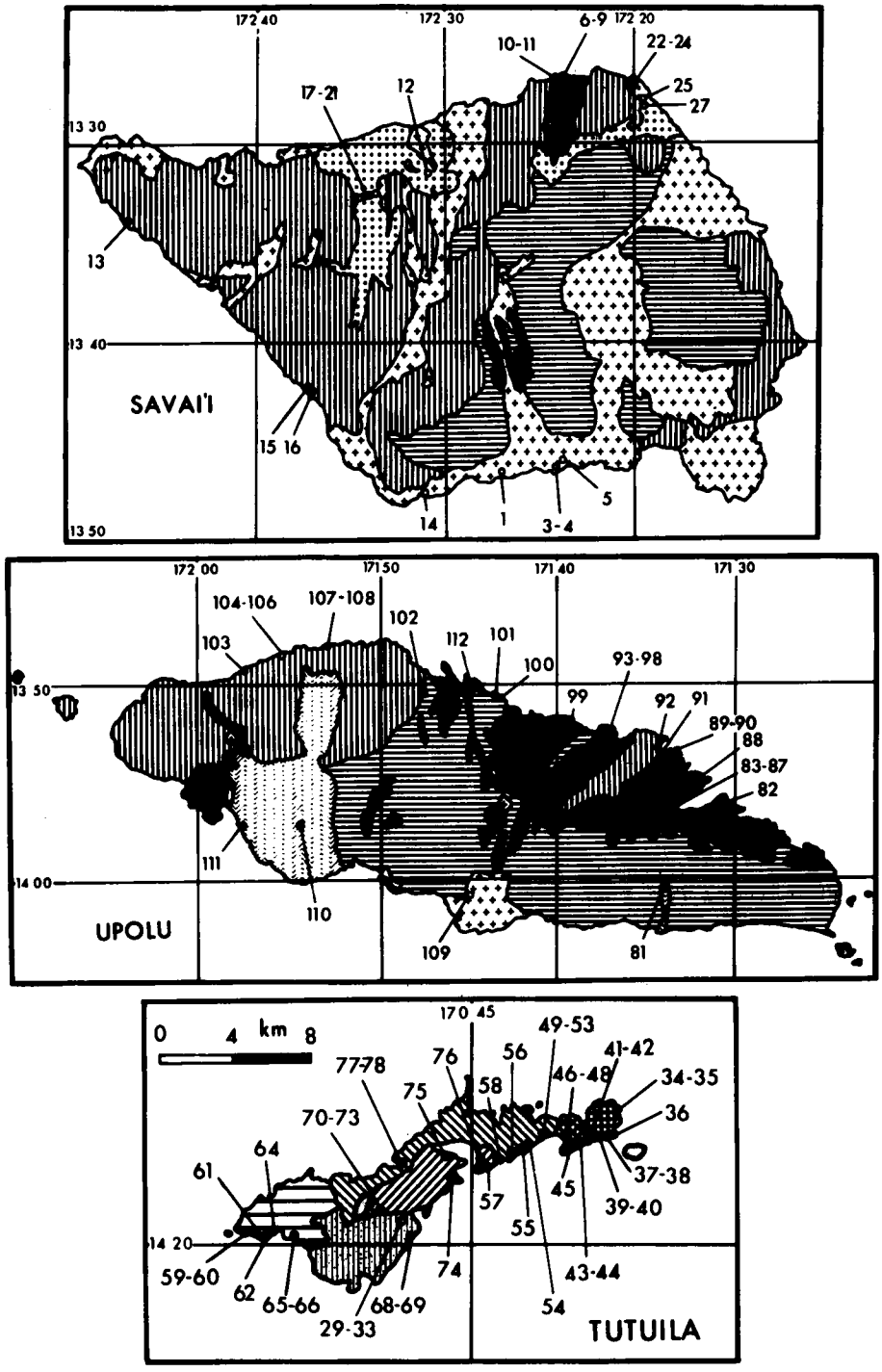



FIG. 1a. Samoa: site localities and geology. Based on Kear & Wood (1959) and Stearns (1944).

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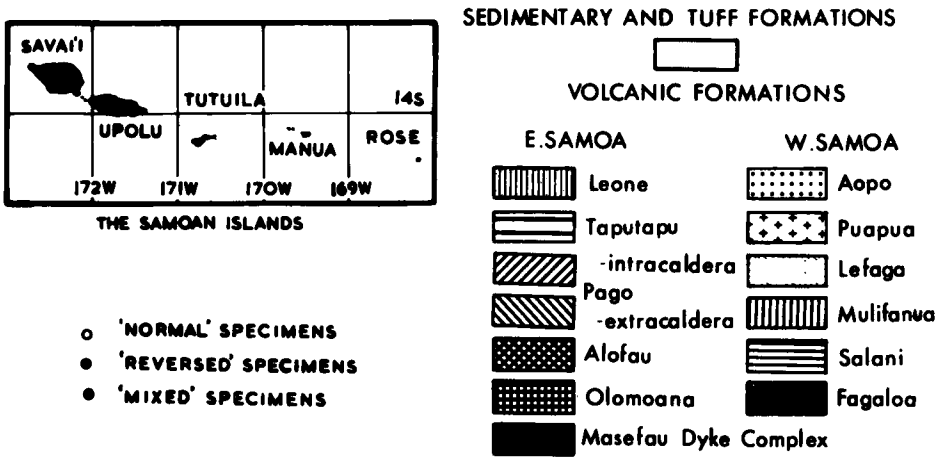



FIG. 1b. Key to Fig. 1a.

3.1. *Geology*

The Samoan Islands (Fig. 1, Table 1) have been visited by several geologists in the last 100 years, but no detailed work was published until Daly (1924) reported his observations in American Samoa. Daly's work was amplified and the succession determined by Stearns (1944), who also made brief notes on Western Samoa formations. Macdonald (1944) described the petrology of the samples collected by Stearns. In Western Samoa the only detailed published work is by Kear & Wood (1959).

On Tutuila, the oldest formation is the Masefau Dyke Complex, which includes lavas as well as dykes, but was not sampled. The Olomoana and Alofau formations are thought to be contemporaneous, although activity ceased first in the Olomoana volcano. The Pago formation overlaps the Alofau lavas, and consists of two units: the extra-caldera unit being separated from the younger intra-caldera unit by a fault zone which is believed to be part of the old caldera wall. The Taputapu formation is probably older than the Pago formation as it is less dissected, but its relationship with the preceding formations is not clear. The lavas of the Leone formation are undoubtedly the youngest igneous rocks on the island as the formation still includes superficial cinder cones, and has only a thin soil cover and no fringing reef. In composition, the predominant rock type on this island is olivine basalt, although rocks varying from picrite to trachyte are found.

The formations on the islands of Upolu and Savai'i were mapped at the same time (Kear & Wood 1959). The oldest formation, the Fagaloa formation, outcrops on both islands, although it is possible that on Upolu the formation can be divided into a younger and older sequence (Kear, personal communication). This formation was subjected to considerable erosion prior to the eruption of the Salani formation which was followed in comparatively quick succession by the Mulifanua, Lefaga and Puapua formations. The youngest formation is the Aopo formation, which consists of two flows, one occurring about 1760 (Thomson 1921) and the other during 1905–11. In composition, all the igneous formations of Western Samoa are predominantly olivine–basalt although the composition ranges from picrite to trachyte.

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The ages of the igneous formations are not known accurately, except for the historic Aopo formation (Savai'i). Earlier formations are correlated on the basis of superposition and the extent of erosion associated with various base levels attributed to eustatic changes of sea level during the Pleistocene. Kear & Wood (1959) concluded, on this basis, that the Fagaloa formation on both Upolu and Savai'i was extruded prior to the penultimate glaciation, and Stearns (1944) suggested that the Pago formation, Tutuila, was the same age as rocks on Upolu which have since been mapped as the Fagaloa formation.

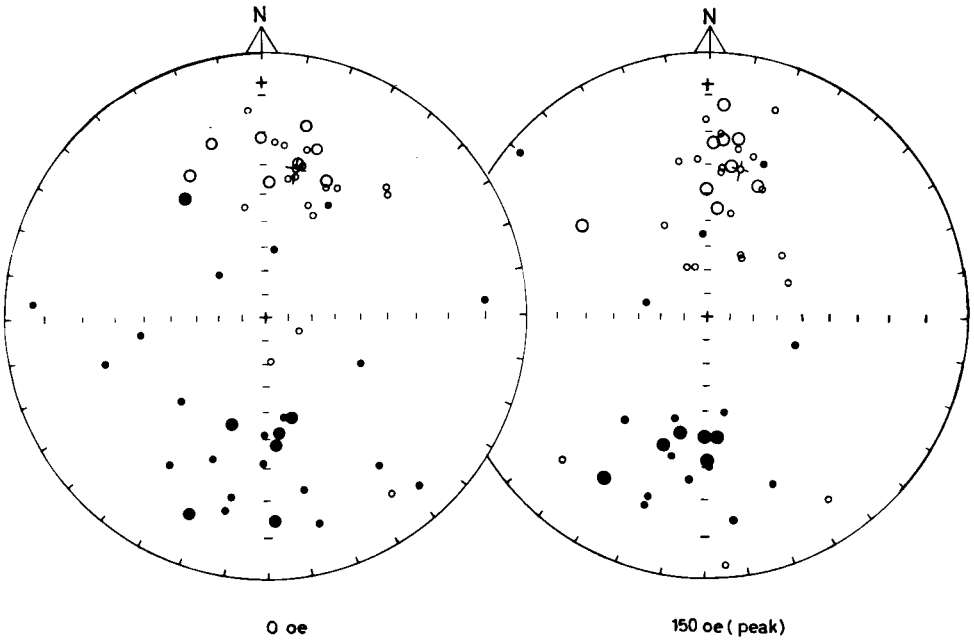



FIG. 2. Site directions on a stereographic projection. Open circles are negative (upward) inclinations, dots are positive (downward) inclinations. Large circles or dots represent significant sites, small symbols represent sites in which only one sample was obtained. Random site directions are not shown. +, axial dipole field. -⊙-, present geomagnetic field.

### 3.2. *Directions of remanence*

Significant site directions, and sites in which only one sample is available, are plotted in Fig. 2 and the formation mean directions, calculated giving equal weight to each site, irrespective of significance, are listed in Table 2. The significant site directions fall into two groups, normal and reversed. All significant sites in Western Samoa are normal, except in the Fagaloa formation on Upolu where both polarities occur, and all significant sites in Eastern Samoa are reversed, except in the Leone formation which are all normal. Sample or specimen directions, although scattered, fall into the same two groups—the polarity being related to the stratigraphical position of the formation (Table 1).

The scatters of site directions behave differently in different formations as the specimens are partially demagnetized. Some formation directions improve their precision, some remain the same while others decrease. Sixteen of the twenty-five sites in which significance can be tested are significant at a 95% level after partial

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**Table 2**

*Mean formation directions in Samoa (calculated giving equal weight to each site)*

| Formation | Number of sites | Number of samples | Number of specimens | Initial | | | | Treated (150 oe) | | | | | Pole | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | *D* | *I* | **R** | *k* | *D* | *I* | **R** | *k* | α | Lat. | Long. |
| Aopo (1905–11) | 5* | 5 | 11 | (13 | −29)* | – | — | (9 | −30)* | — | — | — | (81° N | 64° W)* |
| Aopo (*c.* 1760) | 6* | 6 | 12 | (18 | −23)* | — | — | (10 | −21)* | — | — | — | (80° N | 98° W)* |
| Puapua (Savai'i) | 4 | 5 | 11 | 14 | −31 | 3·86 | 21 | 13 | −33 | 3·89 | 27 | 18 | 77° N | 63° W |
| Puapua (Upolu) | 1 | 1 | 2 | (15 | −24) | — | — | (16 | −25) | — | — | — | (74° N | 83° W) |
| Leone (Tutuila) | 2 | 6 | 12 | 8 | −15 | 1·97 | 32 | 11 | −23 | 1·99 | 96 | 26 | 79° N | 110° E |
| Lefaga (Upolu) | 2 | 2 | 4 | — | — | 0·86 | — | — | — | 1·88 | — | — | — | — |
| Mulifanua (Savai'i) | 2 | 3 | 7 | 10 | −20 | 1·99 | 157 | 4 | −16 | 2·00 | 211 | 17 | 83° N | 135° W |
| Mulifanua (Upolu) | 3 | 6 | 12 | 20 | −39 | 2·68 | 6 | 4 | −33 | 2·96 | 56 | 17 | 84° N | 31° W |
| Salani (Upolu) | 3 | 3 | 6 | — | — | 2·17 | — | 7 | −47 | 2·83 | 12 | 37 | 74° N | 31° E |
| Fagaloa (Savai'i) | 2 | 6 | 9 | 0 | −26 | 1·97 | 30 | — | — | 1·86 | — | — | — | — |
| Fagaloa (Upolu) (*N*) | 6 | 7 | 14 | 23 | −43 | 4·87 | 4 | 7 | −52 | 5·22 | 6 | 29 | 70° N | 27° E |
| Fagaloa (Upolu) (*R*) | 6 | 9 | 20 | 176 | +27 | 5·50 | 10 | 184 | +39 | 4·93 | 5 | 35 | 81° S | 162° E |
| Taputapu (Tutuila) | 5 | 7 | 13 | — | — | 2·82 | — | — | — | 1·30 | — | — | — | — |
| Pago intracaldera | 5 | 9 | 17 | — | — | 2·81 | — | 201 | +31 | 3·81 | 3 | 49 | 69° S | 108° E |
| Pago extracaldera | 5 | 10 | 20 | 185 | +13 | 3·53 | 3 | 204 | +32 | 3·35 | 2 | 63 | 67° S | 110° E |
| Alofau (Tutuila) | 5 | 10 | 20 | 182 | +38 | 4·94 | 71 | 185 | +34 | 4·94 | 67 | 9 | 84° S | 143° E |
| Olomoana (Tutuila) | 5 | 5 | 11 | 190 | +55 | 3·70 | 3 | — | — | 2·23 | — | — | — | — |

*D*, *I* are the declination from true North and the inclination from the horizontal (+down, −up). **R**, *k*, α are the length of the resultant vector, the estimate of precision and the semiangle of a cone of 95% confidence (Fisher 1953). Mean directions, etc., are not given if **R** is not significant at a 95% level (Section 2) and are in parentheses if the significance cannot be tested. The estimates of *k* and α must be treated with caution when the number of sites per formation is less than five. (*N*) normal and (*R*) reversed sites.

* All sites are in the same lava flow.

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demagnetization (Section 2) of which nine have normal polarity (1°, −30°, $\alpha = 14°$) and seven are reversed (191°, +35°, $\alpha = 10°$). These significant sites occur in ten different formations and are considered (Section 3) to be reliable indicators of the direction of the past geomagnetic field in Samoa. Their mean direction (5°, −32°, $\alpha = 8°$) is not significantly different from the present (12°, −30°) or axial dipole field (0°, −27°) in Samoa but is significantly displaced from the inclined dipole field (13°, −28°) in Samoa by 1°. The reversed site directions, considered irrespective of sign, are closer to the present and inclined dipole field than are the normal, although the two mean directions, within the statistical limits, are exactly reversed.

The estimates of precision of the scatter of significant site directions about their mean (Section 2) are given by $k = 20$ (circular standard deviation $\psi_{63} = 18°$) on the Fisher analysis, and $\beta = 25(\psi_{63} = 16°)$ on the Watson & Irving analysis. These dispersion estimates may be compared with (a) the geomagnetic (1960 field) scatter around the latitude (14° S) of Samoa, $\psi_{63} = 21°$, (b) with the average of the geomagnetic scatter around the latitudes 14° N and 14° S, $\psi_{63} = 17{\cdot}4°$, and with (c) the geomagnetic scatter along that part of the latitude of the Samoan Islands which is within the Pacific (as defined by the Andesite Line), $\psi_{63} = 8{\cdot}4°$. It is clear that the palaeomagnetic scatter is similar to the geomagnetic scatter around the complete line of latitude of Samoa, or the average of 14° N and 14° S, and is considerably larger than the scatter observed over the same latitude in the Pacific region alone.

### 3.3. *General magnetic properties*

The intensity of NRM of all specimens was measured and averaged arithmetically to obtain sample intensities. The initial susceptibility, which is the ratio of the intensity of magnetization induced in an external field to the strength of that field

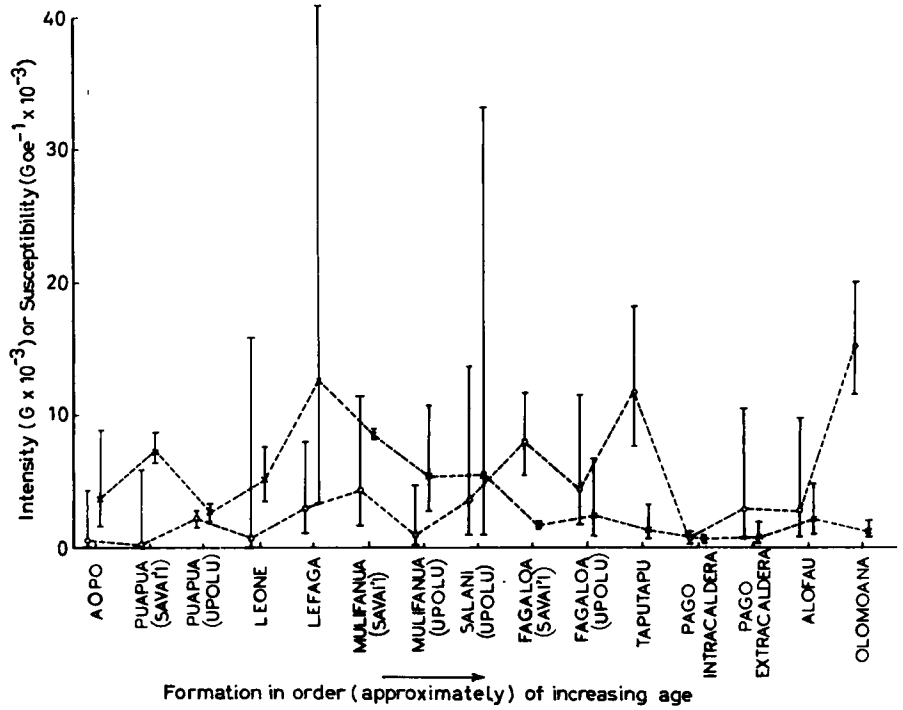



FIG. 3. Average intensity and susceptibility of Samoan formations.

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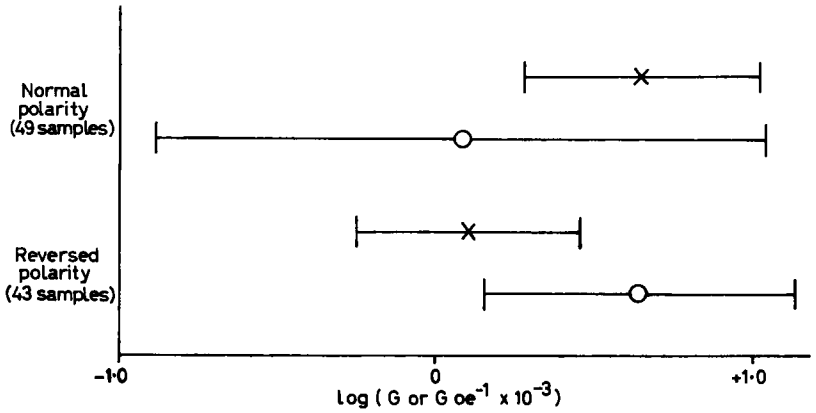



FIG. 4. Average intensity and susceptibility of normal and reversed samples. ×, intensity; ○, susceptibility.

(0·5 oe), was measured of one specimen from each sample and the average value of susceptibility and intensity (Fig. 3) was calculated for each formation from this data, assuming a log–normal distribution. The older formations have a generally higher susceptibility but lower intensity of NRM than the younger formations. This difference is emphasised in the average of intensity and susceptibility for all normal and all reversed samples (Fig. 4); normal samples have an average intensity four times higher and an average susceptibility four times lower than reversed samples.

One specimen from each of three normal and three reversed formations was given an isothermal remanence (Roquet 1947). The maximum remanence, $0{\cdot}28 \pm 0{\cdot}13$ G, and the applied field required to produce it, $455 \pm 253$ oe, are both much lower than that observed in Hawaiian basaltic samples (Tarling 1965) and the coercivity of maximum remanence, $283 \pm 39$ oe, measured in zero field, is also lower.

Four specimens, two normal and two reversed, were given a thermal remanent magnetization (TRM) by heating them to 750 °C in a non-magnetic furnace (Irving *et al.* 1961b) then allowing them to cool in the Earth's field (0·59 oe) with their orientation marks, which were randomly oriented with respect to their field positions, in the magnetic meridian. The directions of TRM were the same, within experimental error, as the applied field suggesting that the specimens are isotropic to TRM and the reversed NRM specimens behaved similarly to the normal specimens in showing no self-reversal tendency. The TRM intensities were higher than the NRM intensity, the ratio NRM/TRM being 0·20, 0·09, 0·01 and 0·03.

### 3.4. *Stability*

In order to obtain an indication of the stability of NRM, forty-five pilot specimens were partially demagnetized in incrementally increasing alternating magnetic fields up to 300 or 450 oe (Section 2). The behaviour of directions was variable (Fig. 5), some remaining constant over a small range of applied field, while others had apparently random directions in successively higher fields. The variability of directional behaviour was reflected in the behaviour of intensity (Fig. 5) in successively higher fields. The intensity of NRM of specimens in which the direction remained constant usually decreased gradually, although the intensity of reversed specimens often increased slightly before the gradual decrease commenced. Other specimens behaved in a variety of ways—some decreased rapidly and remained low, others fluctuated between successively applied fields. This type of behaviour suggests

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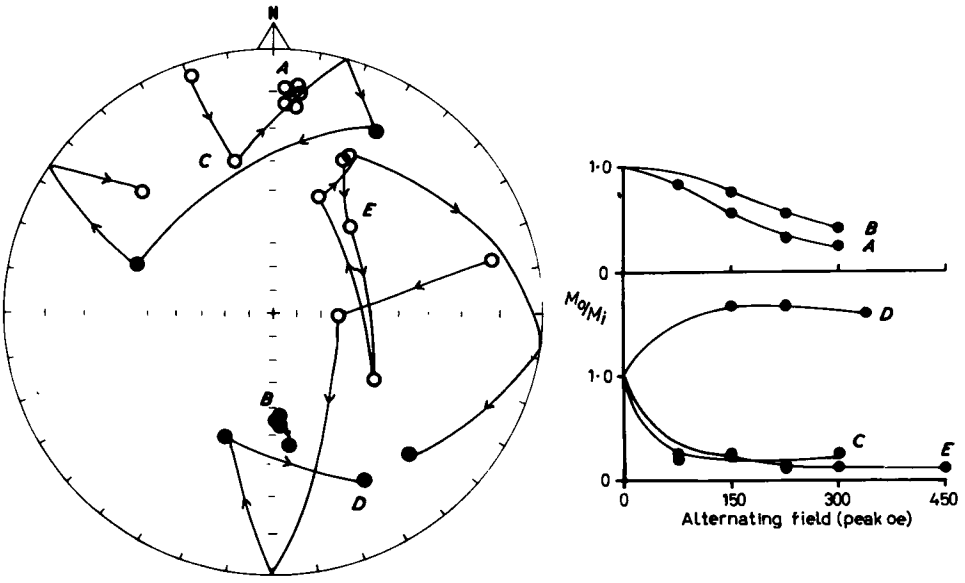



FIG. 5. Alternating field demagnetization of typical Samoan pilot specimens.

A. Aopo (1760), specimen 20a } Typical of stable
B. Pago intracaldera, specimen 50a } specimens.
C. Fagaloa, Savai'i, specimen 10a } Typical of meta- and
D, E. Fagaloa, Upolu, specimens 85b and 101a } unstable specimens.
$M_0/M_i$ = ratio of observed to initial intensity.

a large unstable component is present in many specimens, but some specimens have stable directions over certain ranges of applied field. Examination of the constancy of directions showed that most specimens in which there is some indication of stability in alternating fields retained approximately constant directions over a range which included 150 peak oe, and all other specimens were treated in this field in order to reduce the effect of low coercivity components. However, in view of the general behaviour of pilot specimens, other criteria must be used to distinguish stable material after partial demagnetization.

The consistency of specimen directions in formations older or younger than the Fagaloa (Upolu) formations suggests that the material is sufficiently stable for determination of the polarity but not necessarily the precise direction of the original magnetic field in which the lavas cooled. Only twenty-five site directions could be tested for the similarity of sample directions and only sixteen of these had mean directions significant at a 96% level. It is concluded that only the sixteen significant site directions can be regarded as probably stable.

### 3.5. *Discussion*

The similarity of the between-site scatter of significant site directions with that of the geomagnetic scatter around the latitude of Samoa suggests that the magnitude of secular variation in the Samoan region when considered over approximately two million years is similar to that expected in other parts of the world for similar latitudes, and is much greater than that observed during historical times in this region. This result is consistent with observations in Hawaii (Tarling 1965) but more detailed work is required to distinguish the periodicity of this phenomenon as it is possible

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that short period components of secular variation may be screened out without affecting the magnitude of the variation when observed over a long time.

The presence of reversed magnetization in older formations is consistent with observations in other parts of the world for rocks of similar age (Tarling 1962) and although self-reversal cannot be excluded as a possible explanation, it is more probable that the reversal reflects a reversal of the geomagnetic field approximately one million years ago and may be used for geological correlation. Assuming that a 'normal' zone has not been missed during sampling, then comparison with Hawaiian polarity zones, which have been dated using radioactive isotopes (McDougall & Tarling 1963), suggests that the rocks sampled do not exceed 2·6 m.y. in age, and therefore the overall migration of volcanic activity from E to W is at an average rate not less than 7·7 cm/year and probably similar to the migration rate of 10 cm/year observed in Hawaii (McDougall 1964).

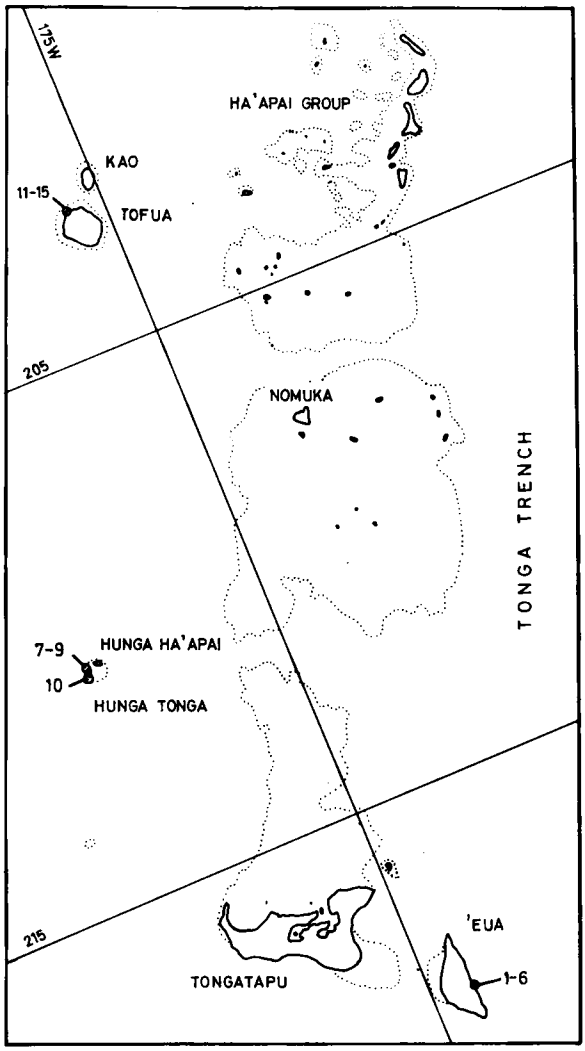



FIG. 6. Tonga: site localities. ....., 100 m depth contour.

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The similarity of the significant site directions to the axial dipole direction is consistent with other stable rocks of similar age in other parts of the Earth. This similarity suggests that there has been no large-scale rotation of the Samoan Islands during the last 2·5 m.y. although the slight divergence of reversed site directions from the axial dipole field could reflect a clockwise rotation of a few degrees.

The time dependence of the susceptibility and intensity of NRM is not understood. The decrease in intensity probably reflects the spectra of relaxation time of TRM while the susceptibility may reflect either variation in iron content, grain size, exsolution, annealing, or other phenomena. The variability of the NRM/TRM ratios is not thought to reflect variations in the strength of the Earth's magnetic field, but is probably attributable to instability of NRM and thermo-chemical changes during laboratory heating and cooling.

## 4. Tonga

The Tongan islands lie in two straight lines on the 'continental' edge of the Tonga Trench (Fig. 6). The western chain rises steeply from the ocean floor and consists of a series of active or quiescent volcanoes which tend to increase in age from N to S. These volcanoes appear to be composed of andesitic lavas (Harker 1891, Hoffmeister 1926, Raitt *et al.* 1955, O. N. Warin, personal communication). The eastern chain, formed by a broad, shallow platform, comprises numerous low-lying calcareous–tuffaceous islands, the age of which is probably Quaternary as the limestones forming Tongatapu, typical of this type, are of this age (Ostergaard 1935). The island of 'Eua, which appears to be a part of the eastern chain, is exceptional as it is high (300 m) compared with other islands of this type, and is geologically more complex, comprising (Hoffmeister 1932) lavas which were faulted and cut by dykes

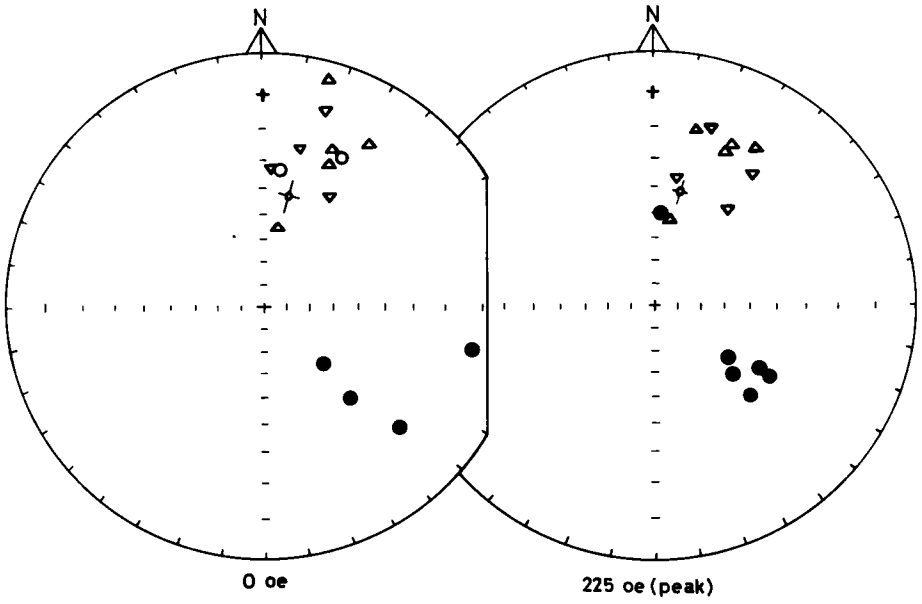



FIG. 7. Tonga: sample directions. ▽, Tofua; △, Hunga Tonga; ○, 'Eua. Solid symbol indicates positive (downward) inclination. +, axial dipole field. ⊹ present geomagnetic field.

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**Table 3**
*Tongan site directions*

| Site | No. of samples | No. of specimens | Initial D | Initial I | Initial R | Initial k | Treated (225 oe) D | Treated (225 oe) I | Treated (225 oe) R | Treated (225 oe) k | Treated (225 oe) α | Pole Lat. | Pole Long. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 'Eua | | | | | | | | | | | | | |
| Northern Dyke | 3 | 9 | — | — | 1·46 | — | — | — | 2·53 | — | — | — | — |
| Central Dyke | 1 | 2 | (134 | +54) | — | — | (122 | +35) | — | — | — | (36° S | 94° W) |
| Main Dyke | 2 | 4 | — | — | 1·94 | — | 129 | +37 | 1·99 | 151 | 20 | 42° S | 94° W) |
| Hunga Tonga | | | | | | | | | | | | | |
| Southern | 1 | 3 | (12 | −25) | — | — | (36 | −25) | — | — | — | (55° N | 92° W) |
| Northern | 3 | 7 | 18 | −27 | 2·91 | 22 | 21 | −32 | 2·91 | 22 | 27 | 70° N | 91° W |
| Tofua | 5 | 11 | 22 | −25 | 4·76 | 17 | 21 | −27 | 4·83 | 24 | 16 | 69° N | 97° W |

Abbreviations as Table 2.

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before being overlain by an upper Eocene limestone and Plio-Pleistocene limestones and tuffs which dip at 10–20° away from the Tonga Trench.

Basaltic andesite samples were obtained from two sites on Hunga Tonga and from one site on Tofua (Fig. 6). On 'Eua, the pre-Eocene lavas were very weathered but six less weathered samples were obtained from three dykes.

### 4.1. *Magnetic properties*

All specimens were treated in a 225 (peak) oe alternating field in order to remove low coercivity components of NRM. The site directions, before and after treatment, are given in Table 3 and the specimen directions are plotted in Fig. 7. The Hunga Tonga and Tofua specimens showed no significant changes in direction after this treatment, but the 'Eua specimen directions became much less scattered. Fifteen pilot specimens were partially demagnetized in peak alternating fields up to 450–750 peak oe (Fig. 8). The direction of specimens from Hunga Tonga and Tofua showed no significant change in any applied field and four of the 'Eua pilot specimens showed only small changes. Two 'Eua specimen directions, however, moved away from the direction of the Earth's field in Tonga and, in alternating fields in excess of 200 peak oe, assumed approximately constant directions similar to the other 'Eua pilot specimens. The intensity of NRM decreased gradually as higher alternating

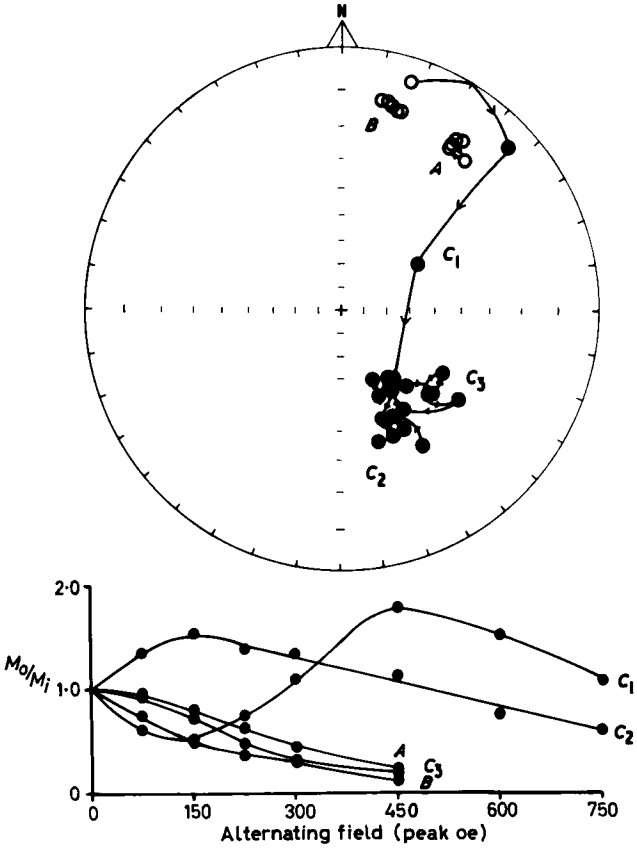



FIG. 8. Alternating magnetic field demagnetization of Tongan pilot specimens. A, Hunga Tonga; B, Tofua; C, 'Eua. $M_0/M_i$ = ratio of observed to initial intensity.

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fields were applied to the Hunga Tonga and Tofua pilot specimens (Fig. 8, curves A and B) but the 'Eua pilot specimen intensities behaved variably (Fig. 8, curve C).

The average values of intensity and initial susceptibility, assuming a log-normal distribution, are given in Table 4 together with the maximum intensity of IRM, its coercivity and the d.c. field required to saturate, which were determined for three specimens. Four specimens were given a TRM and all acquired directions not significantly displaced from the applied field.

**Table 4**

*General Tongan magnetic properties*

| | Average intensity* ($G \times 10^{-3}$) | Average susceptibility* ($G\,oe^{-1} \times 10^{-3}$) | Coercivity of IRM† (oe) | Saturated IRM intensity† (G) | Field required to saturate† (oe) |
|---|---|---|---|---|---|
| 'Eua | 0·9 | 5·5 | 540 | 0·52 | 1 500 |
| Hunga Tonga | 81·0 | 13·7 | 320 | 0·63 | 600 |
| Tofua | 45·0 | 8·9 | 350 | 1·00 | 1 580 |

* Assuming a log-normal distribution.
† One specimen from each group.

4.2. *Reliability*

The directions of specimens from Hunga Tonga and Tofua are considered to be reliable indicators of the past geomagnetic field as the specimens are unweathered and the NRM appears to contain low coercivity components with the same direction as the high coercivity components, both of which are isotropic to TRM. Some 'Eua specimens, however, contain a large low coercivity component directed along the present geomagnetic field in Tonga, the effect of which is only removed after partial demagnetization in alternating fields in excess of 200 oe. The specimens are weathered, but the average directions of NRM at all three sites are consistent after partial demagnetization and are well away from the present geomagnetic field direction. Furthermore one of the specimens tested by partial demagnetization in alternating fields in which the direction remained constant over the complete range from 0 to 450 oe, had a high coercivity of maximum remanence (540 oe). The high coercivity of the 'Eua specimen is suggestive of high stability, and the direction of this specimen agrees with the average mean direction of all other 'Eua specimens after partial demagnetization (120°, +46, $\alpha = 29°$). Therefore it seems that the mean direction of the 'Eua sites, after treatment, can be regarded as stable but as only three sites are involved, it is unlikely that secular changes will have been averaged out in the calculation of the overall mean direction.

4.3. *Discussion*

The mean directions of the Hunga Tonga and Tofua sites are similar to each other, but differ from the fields of the axial dipole (0°, −37°), the inclined geocentric dipole (13°, −36°) and the present geomagnetic field (13°, −41°) in Tonga. However, the difference between the palaeomagnetic directions and the present or dipole fields is well within the statistical estimation of their mean ($\alpha = 33°$ from Hunga Tonga) and, as the sites are probably of similar age, the directions are thought to reflect secular changes of the geomagnetic field during recent geological time.

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The agreement between specimen directions from the three separate dykes on 'Eua suggests that the dykes were all intruded during a short time during which the Earth's magnetic field was reversed. On this assumption, their divergence from the reversed present geomagnetic field (53°) or the axial dipole field (44°) is outside the limits expected from secular variation studies. The palaeomagnetic pole (35° N, 76 °E), calculated on the assumption of an axial dipole field, lies near northern India, which is away from palaeomagnetic poles determined for Mesozoic or Tertiary rocks in other parts of the world. The deviation of the direction seems much more likely to be associated with movement since the intrusion of the dykes as the island is on the extreme edge of the Tonga Trench and the overlying formations are slightly tilted. However, correction for the tilt of the overlying strata (10–20°) makes little difference to the mean direction and if the NRM is associated with a reversed axial geocentric dipole field, either a tilt of some 40° to the SW or some tilting combined with an anticlockwise rotation of up to 60° is required. An anticlockwise rotation could occur between the opposite sides of a left lateral transcurrent fault or an easterly movement of the southern part of the Tongan islands with respect to the northern part. The latter seems unlikely on general grounds and is in the opposite sense to movements suggested by Hess & Maxwell (1949) in the south-west Pacific. The major part of this tectonic activity must have been completed prior to the deposition of the slightly tilted upper Eocene limestones and this suggests that the margin of the Tonga Trench was a region of tectonic instability in pre upper Eocene times, although the smallness of the area that could be sampled prevents determination of the scale of the activity.

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